

Dad, how do you always remember what to purchase from the store?

Oh! How did you make this list?

That is because I always make a list of things I want to purchase.

It is very easy to make a list like this.



What is a list?

A list is a series of data that is written below one another. The data could be either words or numbers.

- A list is easy to read.
- Information can be arranged in any order.

For example: List of Animals arranged as per their size, from smallest to largest

- Cat
- Dog
- Lion
- Cow
- Camel
- Elephant

Data is a collection of information. It can be in the following forms:

- Data in the form of images
- Data in the form of numbers
- Data in the form of text

For example:

I had fun flying a kite - **Data (in the form of text)**

There are 10 apples in the basket - **Data (in the form of numbers)**

Analysis is the process through which we break down a topic or data into smaller parts. This helps us understand the data better.

What is information?

Information is the data that is processed and converted to make it more useful. It can be represented as a list or as a table.

Everywhere we go, we can see examples of data displayed in various formats. We see text, lists, and tables.

1 - List Format

The data is presented in a series, one below the other. This makes it a list.

For example:



a. A menu card



b. A checklist

2 - Tabular format

The data is presented as a table with rows and columns. A tabular format has a few advantages:

- A large amount of data can be presented easily.
- Information can be compared or categorised.

For example:

a. A grade list for students of Class 3

Class 3 - Grade List				
Sr. No.	Name of Students	Math	English	SCIENCE
1	Emma	A	A	B
2	James	B	A	C
3	Noah	B	B	A
4	Mia	A	A	A
5	Sophia	B	B	C
6	David	A	B	B

b. An attendance sheet of a class

Roll No.	Name	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
1	Aarti Sahni	Present	Present	Present	Present	Absent	Absent
2	Bharat Ram	Present	Present	Present	Absent	Present	Present
3	Cheshtha Sharma	Present	Absent	Present	Present	Present	Present
4	Deepak Kohli	Absent	Present	Absent	Present	Present	Absent
5	Jatin Kumar	Present	Absent	Present	Absent	Present	Present
6	Iqbal Siddiqui	Present	Present	Present	Present	Present	Present
7	Keerat Joshi	Present	Absent	Present	Present	Present	Present
8	Mohan Das	Present	Present	Present	Present	Absent	Present
9	Parminder Kaur	Present	Present	Absent	Present	Present	Present
10	Shaurya Dev	Present	Present	Present	Present	Present	Absent

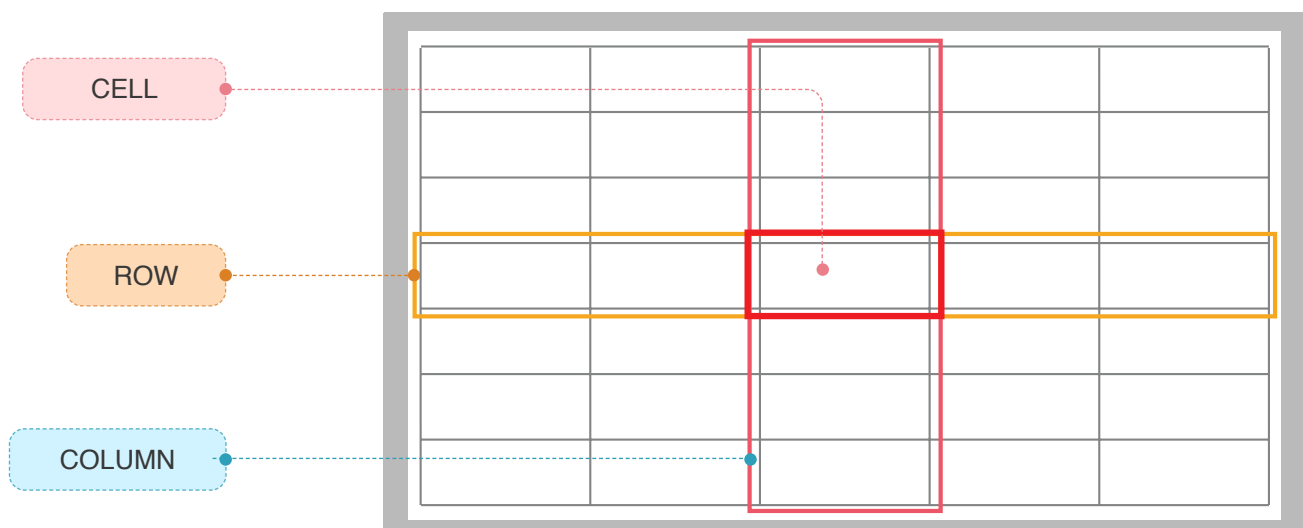
What is a table?

A table is a representation of data using rows, columns and cells.

Cell - The box where data is entered. A cell is formed when a row and column intersect.

Row - A row is a horizontal arrangement of cells.

Column - A column is a vertical arrangement of cells.



Activity 1

Create a table in your notebook and fill data in it.

NOTE - Create a table for the number of classes of each subject in a week. Use the following as titles for the columns:

1. Serial Number

2. Subject Name

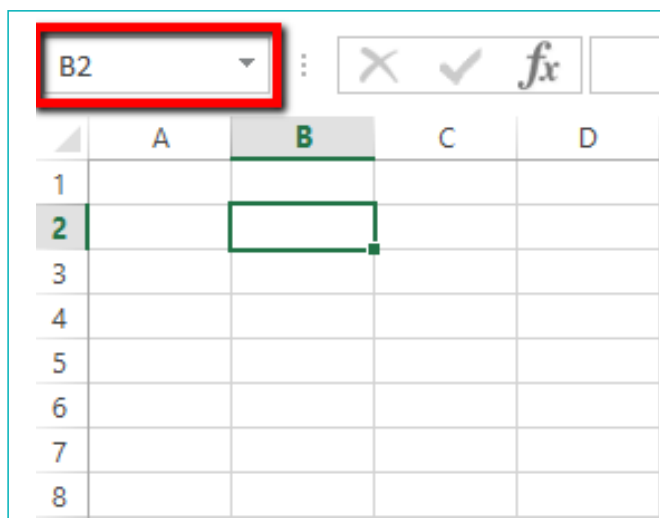
3. Number of classes

What is a spreadsheet?

A spreadsheet is a document that shows data in a table.

- a. It is a grid made of rows and columns where the data is organised.
- b. In a spreadsheet, columns are denoted by alphabets and rows are denoted by numbers.
- c. A cell is denoted by the letter and number of the column and row.

For example: the selected cell is from column B and row 2, so the cell name becomes **B2**.



Spreadsheet

- e. Spreadsheets can also be used to arrange data in a list format.
- f. We use a software called Excel to make spreadsheets.



Excel logo

Exercise

Fill in the blanks

- 1 Data analysis is the process of collecting the _____ and _____ it.
- 2 A table represents data in cells that are arranged in _____ and _____.

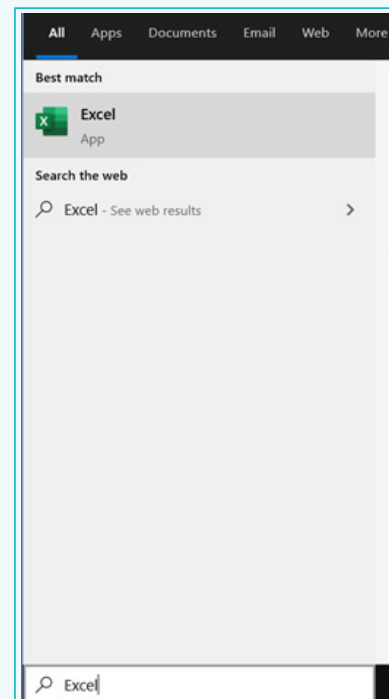
State Whether True or False

- 3 Columns refer to cells arranged horizontally (left to right). _____
- 4 A list helps to arrange data in a sequence typically, one below another. _____

Activity 2

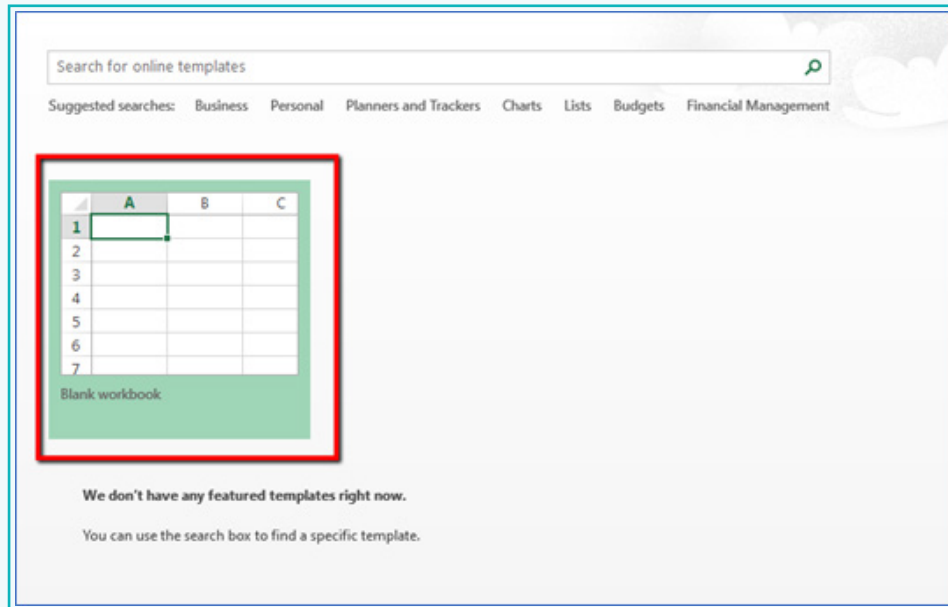
Create a shopping list for vegetables and their quantities in Excel.

- 1 Search for Excel in the taskbar and launch the software.

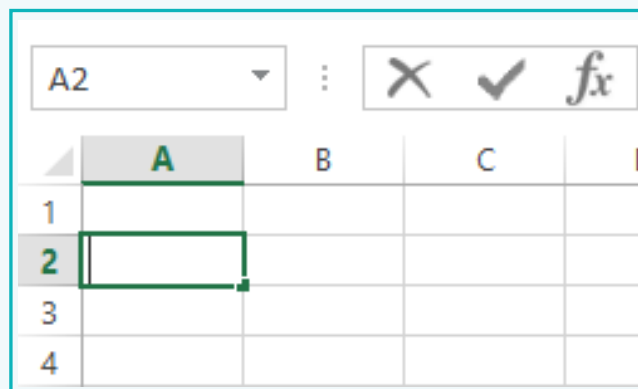


Activity 2

- 2 Click on 'blank workbook' to create a new spreadsheet.

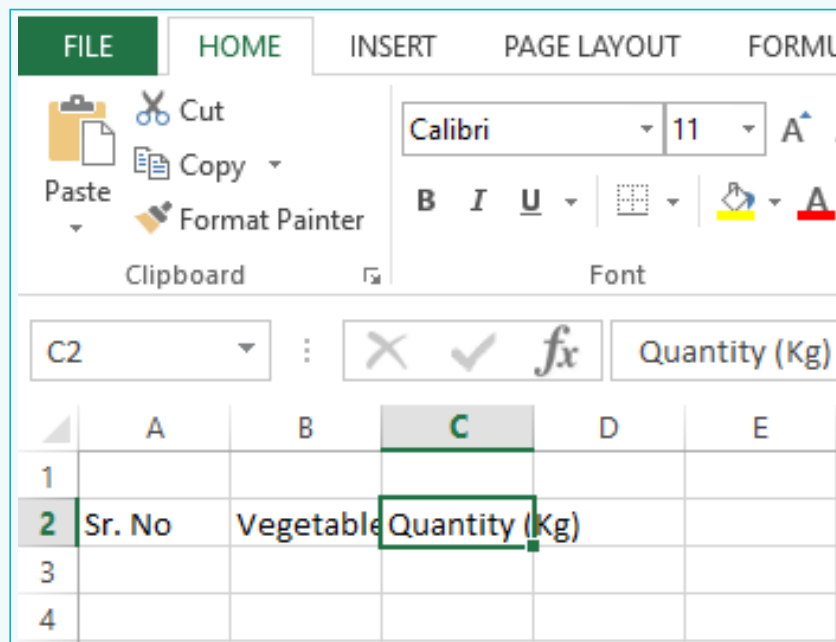


- 3 We can enter data in any row or column. We will use column A and Row 2. To enter data in the cell, double click on the cell.

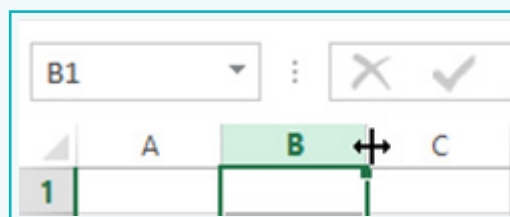


Activity 2

- 4 Generally, a new column is used for each type of data. In this case, we will use 3 types of data:
- a. **Sr. No.** - This stands for Serial Number. We will use this column to keep a count of the entries.
 - b. **Vegetables** - In this column we will write all the names of vegetables.
 - c. **Quantity (Kg.)** - We will use this column to write the quantity of each vegetable.

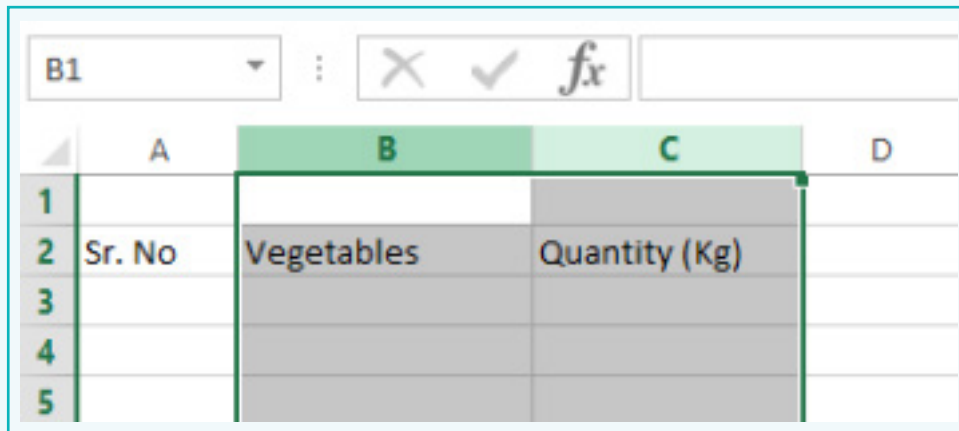


- 5 The width of the cell is too small for the data we have entered. To change the height or width of any row or column, drag the border of that row or column.



Activity 2

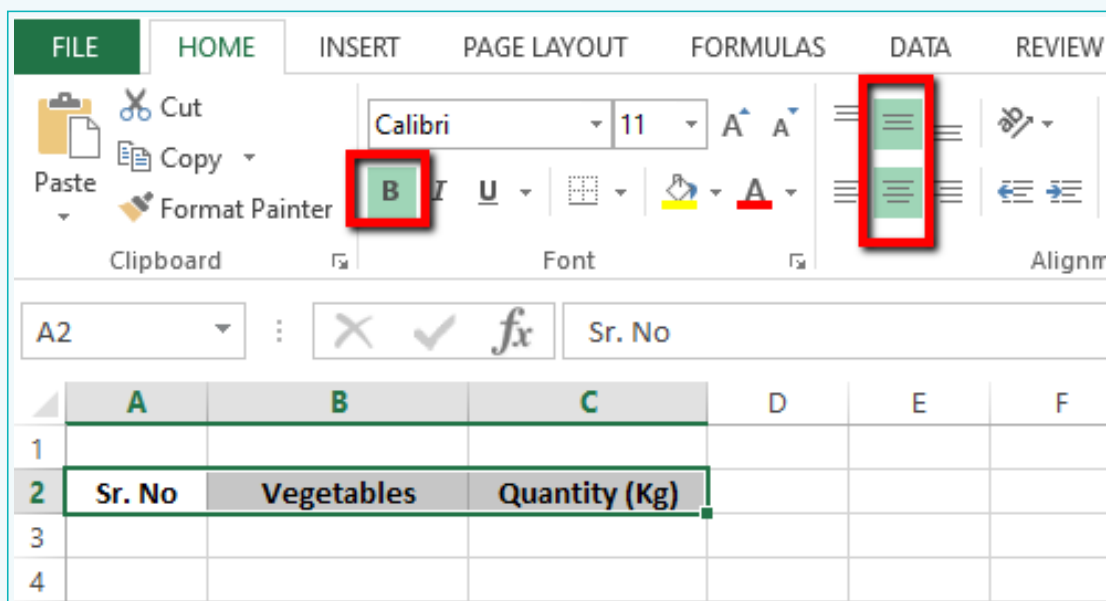
- 6 Increase the size of columns **B** and **C** to fit the data that we have entered.



The screenshot shows a spreadsheet with columns A, B, C, and D. Rows 1 to 5 are visible. Column B is labeled 'Vegetables' and column C is labeled 'Quantity (Kg)'. The cells in columns B and C for rows 2 to 5 are highlighted in grey, indicating they are selected for resizing.

	A	B	C	D
1				
2	Sr. No	Vegetables	Quantity (Kg)	
3				
4				
5				

- 7 Highlight the category titles with **Bold** and **Center Align**.



The screenshot shows the Microsoft Excel ribbon with the 'HOME' tab selected. The 'Font' group is active, showing options for font face (Calibri), size (11), bold (B), italic (I), underline (U), and color (A). The 'Alignment' group is also visible, showing options for text alignment (left, center, right) and orientation. The spreadsheet below shows columns A, B, C, D, E, and F. Rows 1 to 4 are visible. Column B is labeled 'Vegetables' and column C is labeled 'Quantity (Kg)'. The cells in columns B and C for rows 2 to 4 are highlighted in grey, indicating they are selected for formatting.

	A	B	C	D	E	F
1						
2	Sr. No	Vegetables	Quantity (Kg)			
3						
4						

Activity 2

- 8 Rows are generally used for entries. In this case, the entries are vegetable names and their quantities. Enter the data in the **Spreadsheet**.

	A	B	C	D
1				
2	Sr. No	Vegetables	Quantity (Kg)	
3	1	Cabbage	2	
4	2	Carrot	1	
5	3	Potato	2	
6	4	Spinach	1	
7				
8				

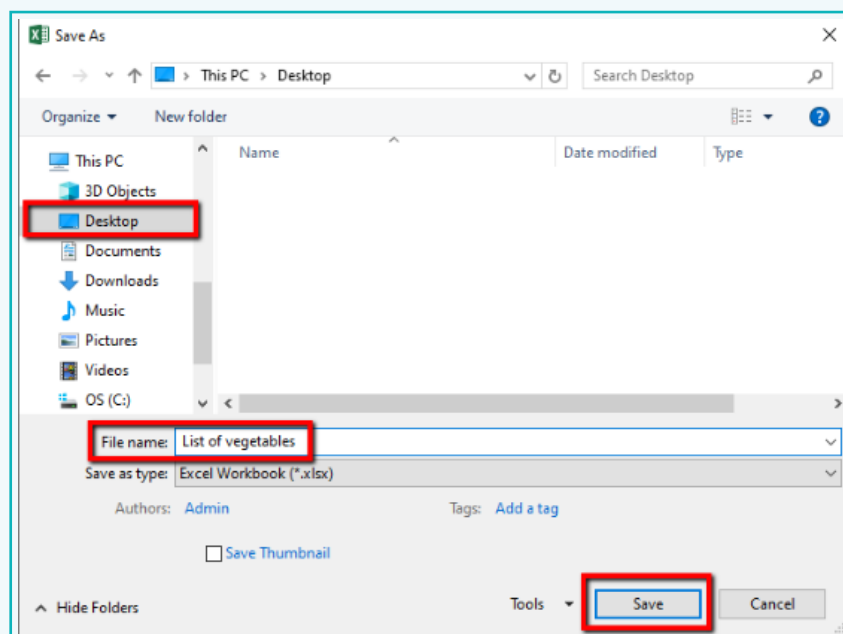
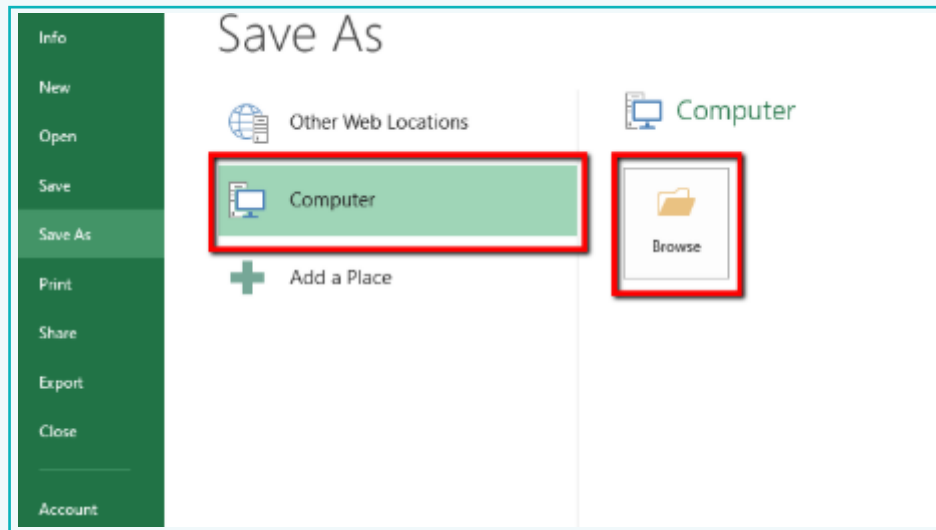
- 9 We will **Center align** all the entries to make the table look uniform. Select all the cells and align the data to the center.

The screenshot shows the Microsoft Excel interface with the 'DATA' tab selected. The 'Center' alignment button in the 'ALIGNMENT' group is highlighted with a red box. Below the ribbon, the spreadsheet shows the data from the previous table with columns A, B, and C highlighted in green.

	A	B	C	D	E	F
1						
2	Sr. No	Vegetables	Quantity (Kg)			
3	1	Cabbage	2			
4	2	Carrot	1			
5	3	Potato	2			
6	4	Spinach	1			
7						
8						

Activity 2

- 10 Save the spreadsheet as 'List of Vegetables'.



Activity 2

Final Output:

	A	B	C	D
1				
2	Sr. No	Vegetables	Quantity (Kg)	
3	1	Cabbage	2	
4	2	Carrot	1	
5	3	Potato	2	
6	4	Spinach	1	

A spreadsheet is one of the easiest ways to arrange data and understand it. It can store, analyse and organise data in tabular and list form.

Spreadsheet is basically a list of items and their details.

Yes, correct!



Tech Flash

- Column** : Vertically arranged cells make a column (top to bottom).
- Row** : Horizontally arranged cells make a row (left to right).
- Table** : A table represents data in cells, arranged in rows and columns.
- Cell** : A cell is the box formed when rows and columns intersect.
- List** : A list is data written in a sequence, one below the other.
- Spreadsheet** : A spreadsheet is a set of rows and columns that stores data.



Future Tech

Machine Learning

Machine Learning is the process through which computers learn to analyse data and make decisions based on the analysis. Over time, their learning automatically improves. It develops computer programs that can access the data and use it to learn by themselves.

